

Magnetic Surveying at the Geophysical Discovery Lab

- Due date: December 8th, 2026

Objectives

Design and implement magnetic surveys over several targets buried beneath Kafadar Commons to identify their locations.

1. Design and establish a survey grid for collecting magnetic data based on given prior information.
2. Design and implement appropriate data acquisition experiments using a cesium vapor magnetometer.
3. Design and implement appropriate data acquisition experiments using a pocket sized magnetometer.
4. Establish a magnetic base station with a proton-precession magnetometer in order to monitor and correct for magnetic drift in your field data.
5. Plot and present your final corrected data in a consistent geospatial format such that anybody could: 1) understand and reproduce your field experiments; and, 2) identify the locations of any geophysical anomalies out in the field.
6. Interpret your geophysical data by identifying the location of the anomalous source bodies and any other features that you were not predicting to find (if any).

Background

Geophysics Discovery Lab:

Many targets have been buried beneath Kafadar Commons as part of the Mines Geophysics Discovery Lab (GDL). These targets contain a broad collection of features representing different categories of geophysical problems, different combinations of physical properties, and different geometries, depths, and depth extents. This includes several magnetic targets with varying shapes and levels of magnetization, including: full UXO grid, manhole cover, long dipping gas-line, “M” with different properties in each segment, overlapping pipes, coffins, metal-cased borehole, and a few rumored hidden targets. In this lab, you will select a feature buried at the Mines GDL, design and set-up an appropriate survey grid, and gather magnetic field data over it. Your instructor will work with you to select the targets and design your survey grids.

In a previous lab we’ve already investigated the archaeology wall and the dipping slab using resistivity.



Magnetometers

Geometrics G-857 Proton-Precession

A scalar magnetometer



[Manual \(https://www.geometrics.com/wp-content/uploads/2018/10/857Manual.pdf\)](https://www.geometrics.com/wp-content/uploads/2018/10/857Manual.pdf)

Geometrics G-858

A scalar magnetometer



Manual (https://geometrics.com/wp-content/uploads/2018/10/858Manual_D2.pdf).

QuSpin QTFM

(Hybrid Scalar + Vector Magnetometer) But also an alkaline optically pumped magnetometer!



User Guide (<https://quspin.com/qt-fm-gen-2/qt-fm-gen-2-user-guide/>)

Geometrics MagArrow

(Scalar cesium-vapor based magnetometer)



Manual (https://www.geometrics.com/wp-content/uploads/2019/10/770-00110-01_F-MagArrow_User_Manual.pdf)

Tasks

Data Collection

Given the above prior information on the Mines GDL

1. Set up a GPS base station for your differential GPS, you should all share this base station.
2. Set up a Magnetics Base station using the Proton-precession magnetometer, apart from your surveys (likely on Stratton commons).
3. Split into teams and choose a target, and a magnetometer. Some good targets are:
 - UXO grid
 - Dipping gas line
 - Manhole Cover
 - Overlying Pipes
 - Borehole (Easy one to visually see as it's at the surface)
 - The "M"
4. Establish a survey grid appropriate for collecting magnetic data over a well-defined target. Items to focus on include grid size, line direction, line spacing, and station (marker) spacing.

Processing

1. Plot the magnetic field as it changes in time as recorded from the magnetic base station during the course of the field survey. Describe the trend and how this will affect your field data.
2. The data from the cesium vapor will likely be provided to you in both a raw, and a corrected form. Use your background knowledge of magnetic methods to describe the corrections made to the magnetic data (drift, as well as the earth's inducing field) (hint: removing the drift in magnetics is done the exact same way as with gravity).
3. You may need to also remove the drift and the background field from your handheld magnetometer measurements as well.
4. Plot the magnetic data collected over each survey grid prior to and after corrections.
 - Present your plots in real-world coordinates
5. Identify the location of any potential anomalies relevant for each survey grid.